Readiness Scores: What Wearables Measure, What Recovery Science Suggests, and How a Readiness Score Should Be Built

A Readiness score isn't a fitness score. It is a score determining "how prepared is my body for training today?". This means it should be driven mostly by short-term recovery state, recent sleep, autonomic signals (involuntary nerve impulses to regulate unconscious bodily functions), and how the last few days of load compare with the person's baseline. Apple's Vitals app is a good example of the general logic here, it looks for overnight metrics that are outside the user's own typical range, not just whether they are outliers in the population. Apple also places Training Load next to those overnight signals so the user can interpret readiness in context. [1]

We need to understand what providers do at the moment so we can compare these to the scientific suggestions and derive the correct algorithm. The broad market pattern is clear even if the exact algorithms are hidden from the public.

WHOOP's public descriptions tie readiness and recovery to heart rate variability (HRV), resting heart rate (RHR), respiratory rate, and sleep, and WHOOP's newer Healthspan feature reinforces the preference for combining sleep and cardio signals rather than relying on just 1 number alone. WHOOP also makes it clear that training demand affects the next day recovery and sleep needs. More recently, WHOOP has expanded its strain score to include muscular load alongside cardio load, which matters because a readiness system is only as good as the load model it reacts to. [2]

Garmin has taken the most explicit "multi-input readiness" approach among the mainstream companies. Garmin's ecosystem describe their Training Readiness as a 1-100 score combining sleep, recovery time, HRV, acute load, and stress, while Body Battery functions as a day-long energy estimate driven by HRV, stress, sleep, and activity. The key design choice is that Garmin puts together overnight recovery with recent training history rather than treating them separately. [3]

Fitbit's older Daily Readiness Score worked in a similar way, using signals like sleep, recent activity, HRV, and RHR to decide whether to push or recover. More recently, Google has shifted its emphasis towards Cardio Load and personalised targets, which tells us that instead of publishing a single readiness formula, Google seems to be leaning towards pairing recovery interpretation with a more explicit load model. [4]

Now we will look into the scientific papers and understand what they suggest about readiness scores. The main theme is that readiness should be based on individual baselines and trends rather than a universal threshold.

This is most clear with HRV. Expert guidance built around athlete monitoring consistently says HRV is only actually useful when interpreted against a personal baseline and over multiple days, not as a single reading. One useful practical idea is that the signal matters when it stays materially below baseline across several days, not because of one weird morning. The same sources also stress that HRV is nonspecific: heavy training, poor sleep, illness, alcohol, travel, and psychological stress can all push it down. [5]

There is also an important measurement caveat. A 2025 study found that pulse rate variability and heart rate variability are not interchangeable during posture transitions, with changes increasing outside stable resting conditions. This is highly relevant for wearables because many devices derive HRV-like signals from PPG (photoplethysmography what a crazy word - it is the optical sensor used by wearables by shining light into your skin) rather than ECG (electrocardiography - This measures the heart's electrical activity directly, usually through electrodes on the skin generally considered the gold standard). The practical implication is, for readiness, your best autonomic input is usually overnight or controlled resting data, not daytime readings that are taken during movement. [6]

Apple's Vitals design actually lines up well with this logic. The company doesn't appear to depend on one metric alone; instead it looks for multiple overnight metrics leaving the user's normal range at the same time, and the algorithm was developed using real-world data from the Apple Heart and Movement Study. That appears to be the right direction for a readiness score, because a single metric can be noisy, but concordant changes across sleep, heart rate, respiratory rate, and related signals are more trusted. [1]

The part we need to add here is which features actually deserve the most influence. The research doesn't give us a perfect readiness formula because readiness is not a clean biological thing you can measure. It's an interpretation made from a few noisy signals. However, the papers do give us a pretty clear order of importance that we can leverage.

The most important input is probably not one single metric but the combination of recent sleep, overnight autonomic recovery, and recent load. HRV is often treated as the headline readiness metric since it reacts quickly to training stress, poor sleep, illness, alcohol, psychological stress and general fatigue. However this is also exactly why it should not be trusted on its own. HRV is sensitive, but not specific. A low HRV can tell us "something is off", but it cannot tell us exactly what is off. This is why papers on athlete monitoring stress that HRV should be interpreted against a personal baseline, ideally over multiple days, rather than as a single morning reading. [7] [8] [9]

In terms of importance, I feel the strongest evidence supports this order:

- Overnight autonomic recovery: This should be one of the biggest blocks. HRV vs personal baseline should be the strongest single physiological input, with RHR as a supporting signal. RHR is less sensitive than HRV but easier for us to interpret. If HRV is down and RHR is up, that is much stronger evidence than either signal alone.
- Sleep adequacy and sleep debt: This should be just as important as HRV, and arguably more important for a regular user. Bad sleep directly affects next-day

performance, mood, recovery and autonomic state. The score should react clearly to short sleep, poor sleep continuity, and any accumulated sleep debt.

- Recent load fit: This is the part that stops readiness from becoming just a sleep or HRV score. A user can sleep well and still not be ready if their recent training load has spiked above what their body is used to. Apple's 7-day versus 28-day Training Load logic and Google's personalised Cardio Load target both support this kind of structure. [10] [11]
- Illness-like anomaly signals: Respiratory rate, temperature, SpO_2 and unusual overnight heart rate shouldn't dominate normal readiness, but they should become very important when multiple of them move in the wrong direction together. This is essentially the Apple Vitals idea: one weird number can be noise, but multiple out-of-range metrics are much more meaningful.
- Subjective context: If we allow users to tag illness, soreness, stress, alcohol, travel, or poor sleep, these should act as modifiers. This is less "scientific looking", but probably makes the score more accurate because the same HRV drop can mean very different things depending on context.

Given this, an idea of weighting for a readiness score could be: sleep adequacy/debt 30%, autonomic recovery 30%, recent load fit 25%, illness/anomaly/context 10%, and data confidence 5%. Within autonomic recovery, HRV should certainly have the biggest share, RHR should be second, and respiratory rate / temperature / SpO_2 should largely act as supporting or warning signals. This isn't to say these exact percentages are proven in any scientific papers (because they aren't), but rather a design choice based on what the evidence consistently points towards.

The score should be strongly personalised and scored on a 0-100 scale. The score should compare the user to themselves, not to population averages. We should use a rolling 21-30 day baseline for HRV, RHR, respiratory rate, temperature and sleep, and compare recent strain using a short acute window against a longer chronic window, for example 7 days versus 28 days.

If correctly implemented, the actual score should react clearly to one bad night, but one bad night should not rewrite the person's baseline. The best behaviour is to dampen the score with rolling baselines and require persistent deviations for larger penalties. For example, one night of low HRV should reduce readiness a bit; three to five days of low HRV, higher RHR and poor sleep should reduce it a lot. That is closer to how physiology actually behaves.

This gives us a score that is not just "did I sleep well?" and not just "is HRV high?". It becomes what a readiness score should be: a personalised estimate of whether the body looks recovered enough for strain today.

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